ALICE L WALTON SCHOOL OF MEDICINE

Opioid Antagonists

Last updated March 3, 2025 · 12 min read · ScholarRx

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Adapted from: Opioids

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Learning Objectives (4) Versions

After completing this brick, you will be able to:

1. Explain opioid tolerance and physical dependence.
2. Explain strategies to mitigate common adverse effects associated with opioids.
3. Recognize the signs and symptoms of opioid overdose.
4. Explain the r reversal agents in an opioid overdose.

CASE CONNECTION

As you give DF a prescription for hydrocodone, his mother asks, "Is this going to be okay, or is he going to get hooked? Is there something else he can take?" DF is a 25-year-old man who has just undergone a particularly difficult extraction of four wisdom teeth, two of which were impacted.

How will you explain your choice of prescription to DF's mother? Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

In the first brick on opioids, we focused on the foundation of the medications (e.g. mechanism of action). In this brick we will take a closer look at the long-term and side effects, as well as exploring the opioid antagonists (i.e., reversal agents).

What is Opioid Tolerance, Dependence, Withdrawal, and Addiction? (LO#1)

Unlike many drugs, opioids may predispose patients to tolerance, dependence, withdrawal, and/ or addiction. This is related to the high concentration of opioid receptors throughout the CNS, and their impact on pain, sedation, reward, and many other pathways. We will take a look at each of these aspects individually.

Tolerance occurs wh t once was and increasing doses of medication are required to produce a similar effect. Over time, opioid receptors can be desensitized or downregulated as a result of frequent opioid agonist use. This downregulation of the receptors reduces signaling outputs on all pathways, namely pain, sedation, respiratory, and reward pathways.

CLINICAL CORRELATION

Consider a patient who has been taking oxycodone for an injury for a few months. Initially, he felt that his pain was controlled. However, a few months later, he noted that the medication was not working as well as it used to. In this patient, opioid receptors have been downregulated, and the body has initiated compensatory mechanisms within the pain pathway, and, ultimately, he has developed tolerance to his current dosage of oxycodone. As a result, to achieve the same effect as when he initially started the medication, he requires a higher dosage. Tolerance can occur as early as two weeks following consistent use of opioids.

Dependence is what happens when the body has adapted to a drug and the patient needs the drug to function at their baseline or normal state. When the drug is no longer ingested, a set of symptoms called withdrawal symptoms (discussed below) may occur. As we discussed above, with frequent opioid use receptors become desensitized or downregulated, and the body will compensate based on this downregulation. If the drug is abruptly taken away, receptors can be upregulated causing an opposite effect. For instance, a patient taking daily opioids for chronic pain th t of their medication can experience hyper e avoided by tapering (slowly decreasing the drug dose), or, in more severe cases, with the use of drugs like buprenorphine or methadone.

Define dependence

Withdrawal is closely related to dependence and mechanistically is the same. It is the body's response to abruptly stopping a drug. Withdrawal can be physical and psychological. With the wide distribution of opioid receptors throughout the CNS, physical symptoms can include tachycardia, hypertension, dilated pupils, shakiness, and muscle pain. Psychological symptoms can include anxiety, agitation, insomnia, mood changes, or depression. The physiological basis of withdrawal is complex but seems to involve activation of the locus coeruleus in the pons of the brainstem, normally involved in stress and panic responses.

Define withdrawal

Addiction, according to the National Institute on Drug Abuse, is "a chronic relapsing brain disease that is characterized by compulsive drug seeking and use, despite harmful consequences." In other words, whereas withdrawal is largely a physiological response to drug cessation, addiction is a complex medical and psychological compulsion to find and use the drug, despite possible harm, and sometimes illegally.

Patients who take opioids can become tolerant and develop dependence and addiction, but dependence does not usually lead to addiction. Between 20-30% of patients prescribed opioids for chronic pain misuse the medication as a result of dependence and addiction. Also, 8-12% of patients prescribed opioids may develop clinical opioid use disorder. With these statistics in mind, it is important to be good stewards of opioids by only prescribing the shortest duration and minimum amount of medication needed to treat the condition.

CLINICAL CORRELATION

Criteria for opioid use disorder can be found in the *Diagnostic and Statistical Manual of Mental Disorders* (fifth edition). Symptoms include but are not limited to craving or strong desire, recurrent use in situations that are physically hazardous, tolerance as defined by the need for markedly increased amounts to achieve intoxication or desired effect or markedly diminished effect with continued use of the same amount of drug, withdrawal symptoms, or using a drug or similar substance to relieve or avoid withdrawal symptoms.

What is tolerance to a medication?

What are the adverse effects of opioids? (LO#2)

Opioids are effective at relieving pain. However, they can be harmful if used in the wrong context. Their adverse effects range from tolerable to lethal, depending on the drug and the dose. Recall in the first opioid brick, we discussed potency differences between different opioids. For example, fentanyl, a very potent opioid, has a higher potential for adverse effects because of its high potency. In addition to how much drug a patient takes, drug levels are also affected by how the drug is metabolized. As mentioned in the previous brick, many opioids are hepatically metabolized by the CYP system and alterations to this system, like genetic polymorphisms, can contribute to the adverse effects.

Non-Lethal Adverse Effects

Among the many non-lethal side effects of opioids, the most common are constipation, nausea, sedation, dizziness, and physical addiction and dependence. Opioids reduce bowel motility. Reduction in gastric motility may lead to constipation and induce feelings of nausea or vomiting. Miosis (pinpoint pupils) due to pupillary constriction may also occur. Some of these adverse effects, such as sedation and dizziness, will be mitigated as the patient becomes adjusted to the dose. On the other hand, to avoid dependence/ addiction it is important to use the lowest dose for the shortest amount of time to treat the patient's pain.

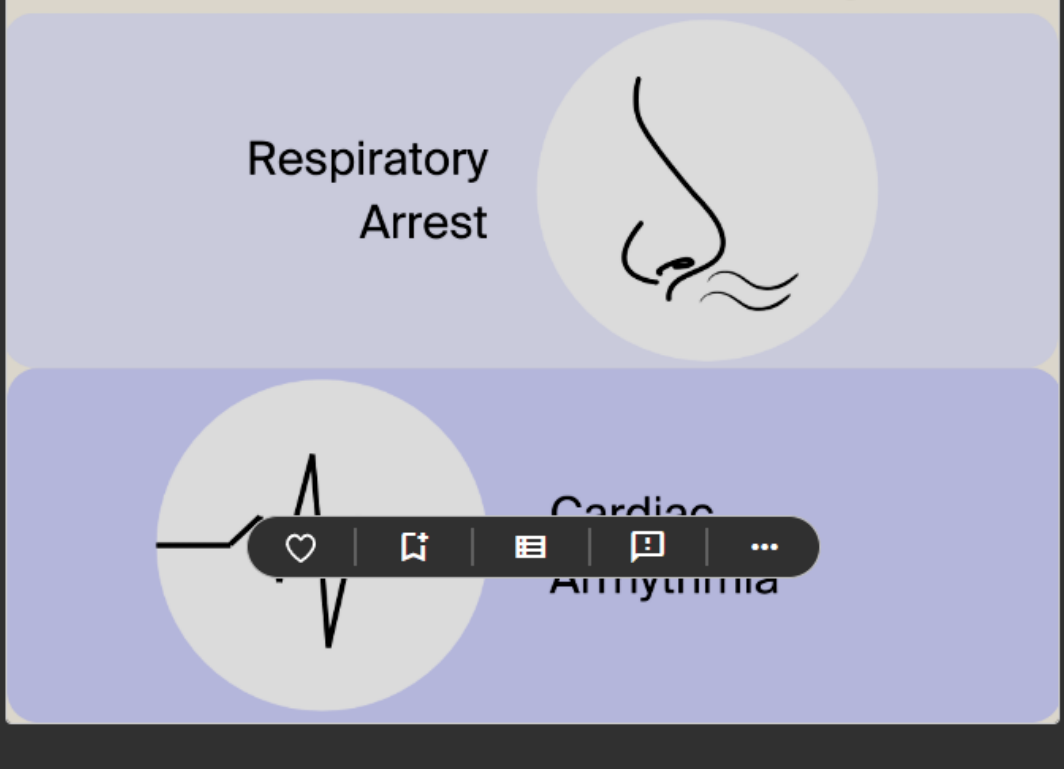
CLINICAL CORRELATION

Opioid induced constipation (OIC) is a common, and concerning, adverse effect for many patients. When prescribing an opioid it is common practice to suggest a bowel regimen to avoid OIC. This would generally include a stool softener (e.g., docusate), a laxativ / tolerated by the patient.

Secondary adverse effects from the abuse of opioids are also possible. Infections can occur when opioids are used intravenously outside of the medical care system, as with heroin or fentanyl. Needle sharing among IV drug users exposes them to bacterial, viral, and fungal infections and can cause heart valve infections or systemic bacterial infection (sepsis) that can sometimes be fatal.

Lethal Adverse Effects

The most common lethal side effects of opioids are respiratory arrest and cardiac arrhythmias.



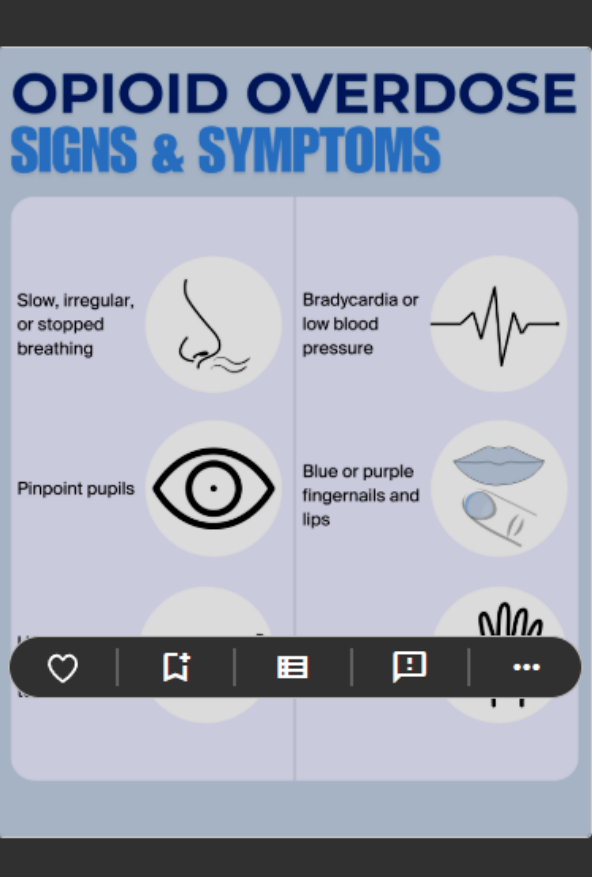
Opioids reduce respiratory drive via binding to the μ receptors. The mechanism is unclear, but it is known that the respiratory rhythm within the brainstem is decreased by the inhibitory effects of opioids. In addition, opioids blunt the brain's chemosensitivity to high blood pCO_2 (partial pressure of carbon dioxide) and hypoxia. These blood abnormalities would normally cause hyperventilation, but this response is lessened when opioids are present. Opioids can also reduce the strength and central drive to respiratory muscles. The net result is that breathing slows and becomes shallower, ventilation drops, and the arterial pCO_2 rises, leading to somnolence and, eventually, respiratory arrest and death.

Certain opioids, such as methadone, influence the cardiac muscle and may prolong QTc. QTc is a measure on an electrocardiogram (ECG) of the length between Q-wave polarization and T-wave repolarization. Having a prolonged QTc can lead to ventricular arrhythmias. Thus, before a patient starts taking methadone, an ECG must be performed.

What are of opioids?

What does opioid overdose look like? (LO#3)

As discussed above in the lethal adverse effects section, opioid overdose generally involves severe respiratory depression and/or severe bradycardia. The patient may present in cardiac arrest and require Advanced Cardiac Life Support. The diagram below shows some of the most common signs of opioid overdose.


OPIOID OVERDOSE
SIGNS & SYMPTOMS
Slow, irregular, or stopped breathing
Bradycardia or low blood pressure
Pinpoint pupils
Blue or purple fingernails and lips

Name three signs of opioid overdose.

How do opioid antagonists work? (LO#4)

Recall our discussion from the first opioid brick on full and partial agonists. In contrast to those agents, opioid antagonists fully bind and inhibit opioid receptors. If an opioid receptor is already bound by an agonist, the opioid antagonist can displace the agonist and reverse the effects. The classic example is the opioid antagonist naloxone. It antagonizes all three receptor types and can reverse the detrimental effects of respiratory depression. See the video below for an animation of opioid antagonists. Note, the duration of action of the opioid antagonis agonists, and often repeat doses of antagonists, like naloxone, are required in overdose situations.

To limit the symptoms of opioid withdrawal for patients who are discontinuing opioids, a tapering strategy or the partial agonist buprenorphine may be used. Another option is methadone, which has much less severe withdrawal symptoms due to the ultra-long half-life of the medication.

such as Naloxone, is given, it has a higher affinity for the opiate receptor and as a result, it

CLINICAL CORRELATION

Naloxone is a life saving opioid antagonist that is carried by all first-responders and now widely available over the counter in many states across the country. Given that naloxone antagonizes all opioid receptors after administration, it can cause immediate withdraw ve and agitated following use.

What is the primary use of naloxone in opioid overdose?

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to DF, how do you explain your choice of prescription to his mother?

You think these questions are reasonable and wise, especially with contemporary concerns about opioid addiction. You say, "You're asking the right questions. First, I have a sense that DF will have a great deal of pain over the next day or two, so he is going to need a medication like hydrocodone to treat it. I'd also like him to start taking a non-narcotic medicine. This will give him even more pain relief, and as soon as he is having less pain, he can stop the hydrocodone and just take the other medicine. By keeping the amount of opioid pain medication to a minimum and using it for the s while minimizing the chance of becoming dependent on it. This prescription is only for 12 pills—just enough to get him through the painful time. You can also do other things to help him have less pain, like applying ice. It's old-fashioned, but it really will help. DF, I'm counting on you to know when to stop the hydrocodone. To be clear, please don't take it if you don't need it to treat the pain." DF and his mother appreciate your detailed explanation and head home with ice bags on both of his cheeks.

Summary

- Opioids work by binding to opioid receptors in the CNS, affecting pain, sedation, reward, and other pathways. This can lead to tolerance, dependence, withdrawal, and addiction.
- Tolerance to opioids can develop as early as two weeks after consistent use, and involves the desensitization or downregulation of opioid receptors.
- Dependence on opioids occurs when the body adapts to the drug and requires it to function normally. Abrupt cessation can lead to withdrawal symptoms.
- Addiction is a chronic relapsing brain disease characterized by compulsive drug seeking and use, despite harmful consequences.
- Adverse effects of opioids range from constipation, nausea, sedation, and dizziness to lethal effects like respiratory arrest and cardiac arrhythmias.
- Signs and symptoms of an opioid overdose include slow, irregular, or stopped breathing, pinpoint pupils, unresponsiveness, bradycardia, blue or purple fingernails and lips, and pale/ clammy skin.
- Opioid antagonists, like naloxone, can reverse the effects of opioids by fully binding and inhibiting opioid receptors.

Review Questions

1. A man is brought by police to the emergency department after being found on the street. When physicians examine him, they notice that the patient has pinpoint pupils, bradycardia, slurred speech, and low blood pressure. A diagnosis of opioid intoxication is made. What medication should be administered to reverse his symptoms?

A. ○ Buprenorphine

B. ○ Methadone

C. ○ Morphine

D. ● Naloxone

Hide Explanation

The correct answer is the naloxone (D). Buprenorphine (A) is only given when a patient is currently in withdrawal from opioids. Morphine (C) is a μ opioid agonist, which would further worsen opioid intoxication and potentially lead to respiratory failure. Methadone (B) is also an opioid agonist used for maintenance treatment of opioid use disorder.

2. __________ is when an opioid is no longer as effective as it once was, and increasing doses a similar effect

A. ○ Addiction

B. ○ Dependence

C. ● Tolerance

D. ○ Withdrawal

Hide Explanation

The correct answer is tolerance (C). Over time, opioid receptors can be desensitized or downregulated as a result of frequent opioid agonist use. Increasing doses of the medication are required to produce the same effects. Addiction (A) is a chronic relapsing brain disease that is characterized by compulsive drug seeking and use, despite harmful consequences. Dependence (B) is what happens when the body has adapted to a drug and the patient needs the drug to function at their baseline or normal state. Withdrawal (D) is the body's response to abruptly stopping a drug.

3. What must be performed before starting methadone?

A. ● ECG

B. ○ Blood test

C. ○ Spirometry

D. ○ MRI

Hide Explanation

The correct answer is ECG (A). Methadone can prolong QTc, which may lead to ventricular arrhythmias, so an ECG is necessary before starting methadone. Blood test (B), spirometry (C), and MRI (D) are not required for assessing the risk of QTc prolongation.

4. Which is a common sign of opioid overdose?

A. ○ Blue or purple fingers

B. ○ Tachycardia

C. ● Pinpoint pupils

D. ○ Hypertension

Hide Explanation

The correct answer is pinpoint pupils (C). Pinpoint pupils are a hallmark sign of opioid overdose. Blue or purple fingers (A) is incorrect as it is the fingernails or lips that will turn blue or purple. Tachycardia (B) and hypertension (D) are signs of opioid withdrawal, rather than overdose.

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How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain opioid tolerance and physical dependence.~~

02 ☑ ~~Explain strategies to mitigate common adverse effects associated with opioids.~~

03 ☑ ~~Recognize the signs and symptoms of opioid overdose.~~

04 ☑ ~~Explain the mechanism of action of agents used as reversal agents in an opioid overdose.~~

Objective Confidence: 4/4